

Perception-Aware Model Predictive Control for Constrained Systems

Chance-Constrained MPC Strategy Accounting for Action-Dependent Perception Quality in Unknown Environments

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Introduction & Interplay of Perception and Control

- Autonomous systems operate under dynamic and unknown environmental constraints.
- Sensor quality (field of view, range, focus, beamforming) depends directly on control actions.
- Traditional MPC treats perception as independent of control, causing conservative trajectories.
- PAC-MPC explicitly couples control decisions with perception quality to optimize safety and motion.

flowchart LR

A[Control Actions u_k] --> B[System State $x_{s,k}$]

B --> C[Sensor Measurement $y_{e,k}$]

C --> D[Environment Estimator]

D --> E[Uncertainty Covariance $\Sigma_{e,k}$]

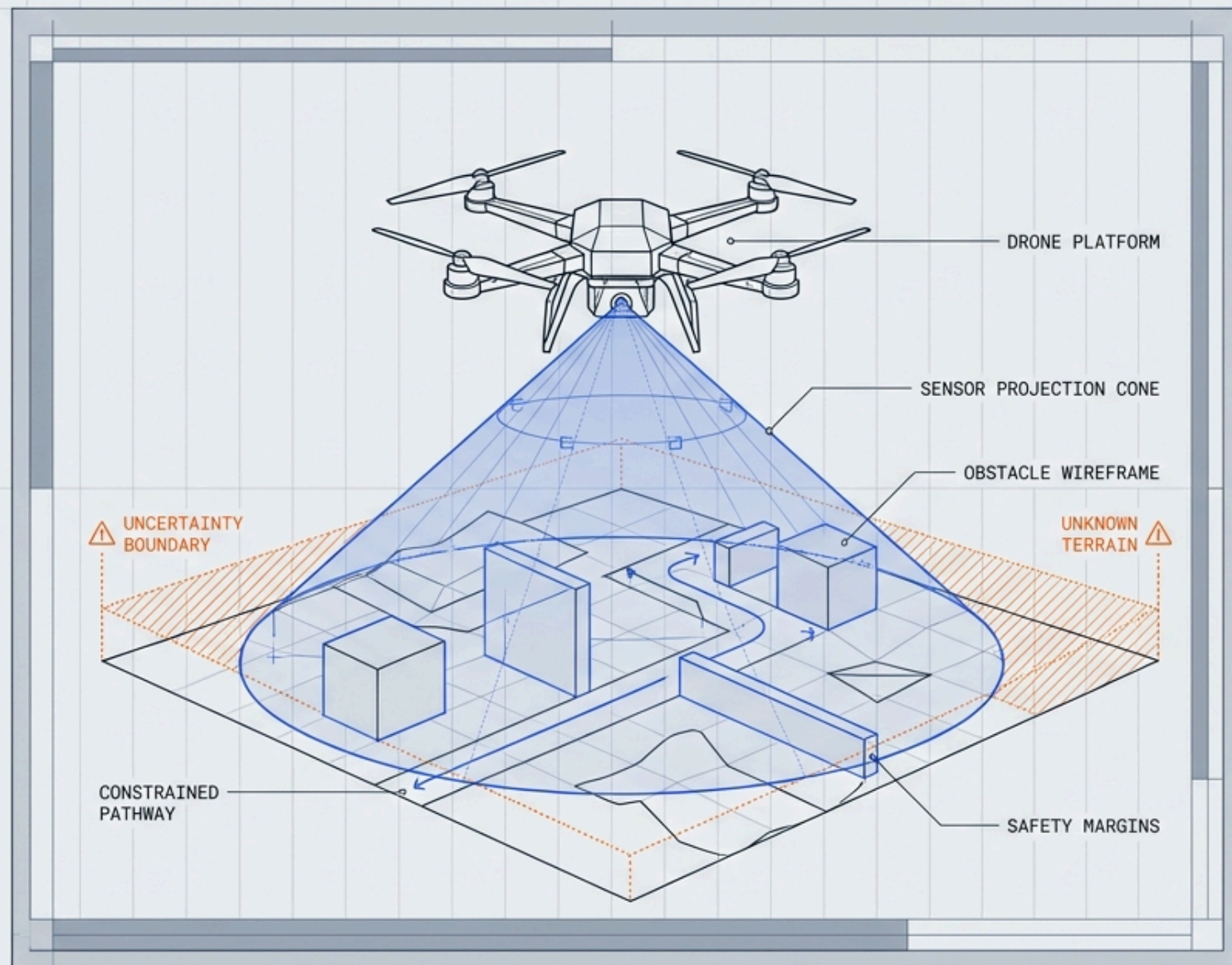
E --> F[Chance Constraints]

F --> A

Perception-Aware Model Predictive Control

Navigating Constrained
& Unknown Environments

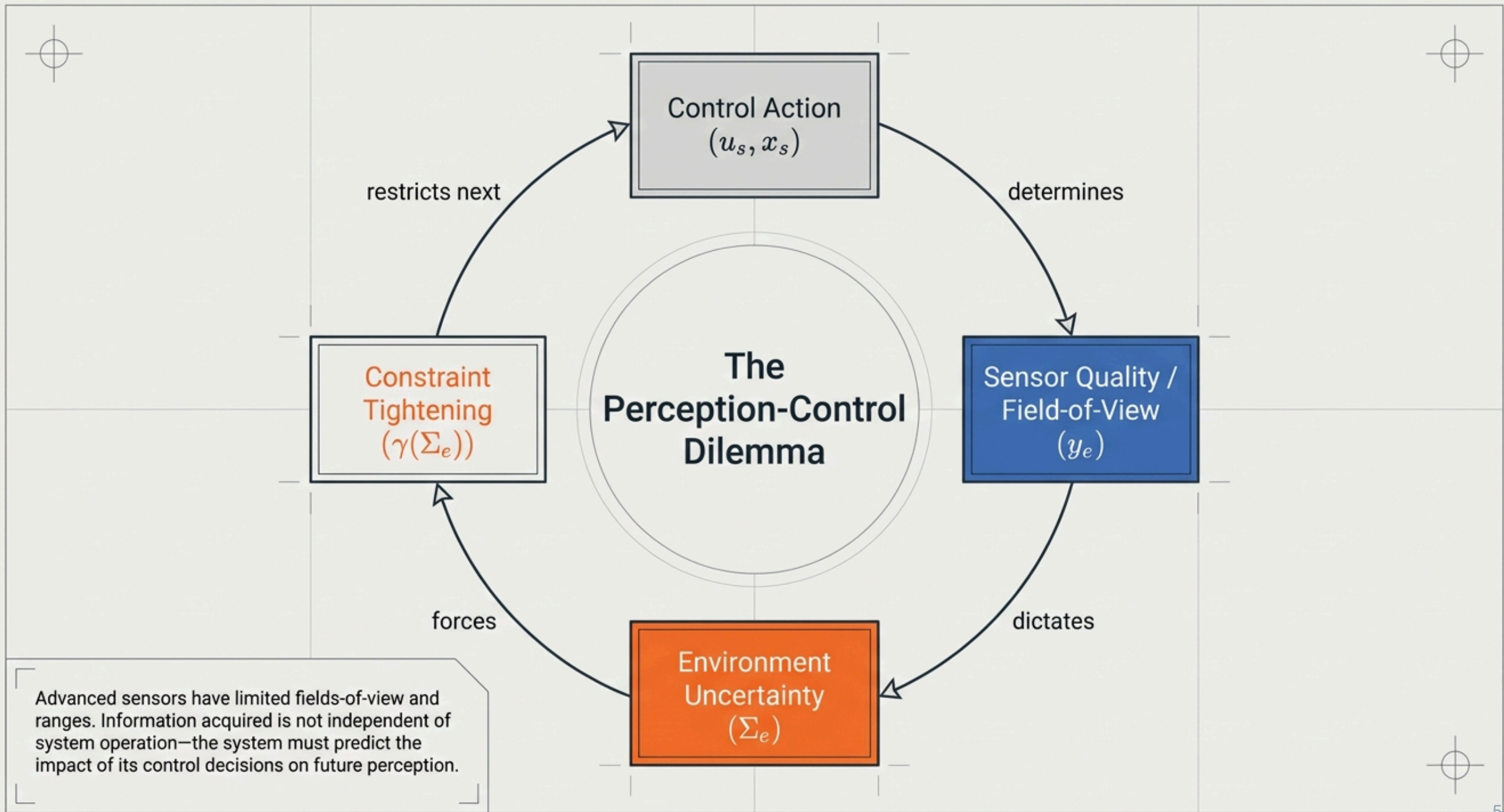
Based on research by Mitsubishi
Electric Research Laboratories
(MERL) & UC Berkeley
(Bonzanini, Mesbah, Di Cairano)



System & Environment Dynamics Modeling

- Known discrete-time system dynamics governed by $x_{\{s,k+1\}} = f_s(x_{\{s,k\}}, u_{\{s,k\}})$ with state/input constraints.
- Exogenous environment state $x_{\{e,k+1\}} = f_e(x_{\{e,k\}}, \psi_k)$ represents static or moving obstacles.
- Environment measurement $y_{\{e,k\}} = q_e(x_{\{e,k\}}, \zeta_k, x_{\{s,k\}}, u_{\{s,k\}})$ depends explicitly on system state and input.
- Individual chance constraints enforce probabilistic safety bounds with violation tolerance ϵ_l .

Model Component	Mathematical Formulation	Key Feature / Coupling
System Dynamics	$x_{\{s,k+1\}} = f_s(x_{\{s,k\}}, u_{\{s,k\}})$	Known deterministic dynamics with state set X and input set U
Environment	$x_{\{e,k+1\}} = f_e(x_{\{e,k\}}, \psi_k)$	Models static or moving obstacles with



Environment Estimation & Predictor Design

- A fixed estimator updates environment mean $\mu_{e,k}$ and covariance $\Sigma_{e,k}$ based on sensor measurements.
- Future measurements y_e are unavailable during horizon planning, necessitating a dedicated predictor.
- The predictor computes predicted moments $(\hat{\mu}_e, \hat{\Sigma}_e)$ using measurement prediction error covariance Σ_y .
- Probabilistic chance constraints are reformulated into deterministic tightened constraints with backoff $\bar{\gamma}(\hat{\Sigma}_e)$.

APPROACH	ENVIRONMENT KNOWLEDGE	CONSTRAINT HANDLING	PERCEPTION INTEGRATION
Standard Deterministic MPC	Assumes Known	Rigid	None
Standard Stochastic MPC	Unknown	Probabilistic	Passive (Reacts to uncertainty)
Perception-Aware Chance-Constrained MPC (PAC-MPC)	Unknown	Probabilistic & Dynamic	Active (Predicts and manipulates uncertainty)



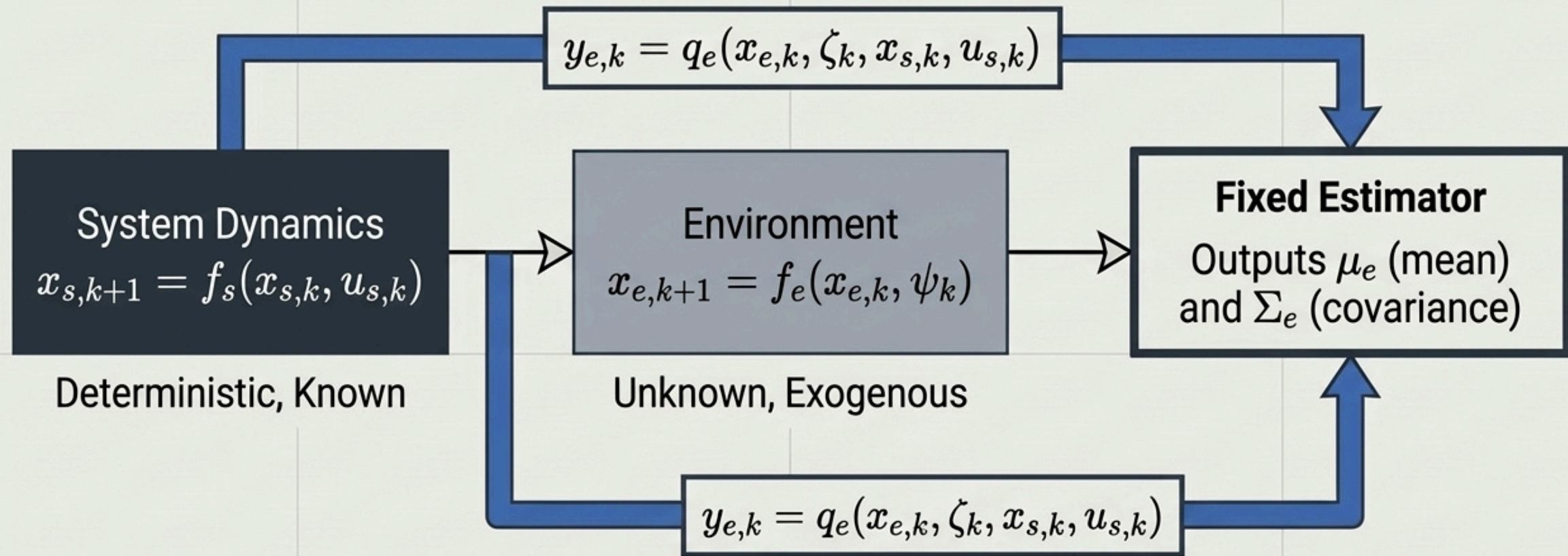
The **PAC-MPC Advantage**: Actively predicting how physical movement alters perception quality, allowing the system to gather information to prevent constraint violation.

PAC-MPC Optimization Problem Setup

- Solves a finite-horizon optimal control problem over prediction horizon N at each sampling step.
- Augmented state $\xi = (x_s, \Sigma_e)$ stabilizes system state to target setpoint and covariance to steady state.
- Cost function combines control stage/terminal costs (l_c, F_c) and perception stage/terminal costs (l_p, F_p).
- Deterministic tightened constraints $h_s \leq x_{s,j|k} + [\gamma(\hat{M}_{e,j|k})] \leq h_b$ enforced along horizon.

graph TD

```
Cost[Min  $V_N$  = Terminal Cost + Stage Costs] --> ControlCost[Control Cost:  $\|x_s - r_x\|_Q^2 + \|u_s - r_u\|_R^2$ ]  
Cost --> PerceptionCost[Perception Cost:  $\|\Sigma_e - \Sigma_r\|_S^2$ ]  
Cost --> Constraints[Constraints]  
Constraints --> Dyn[System & Predictor Dynamics]  
Constraints --> Bounds[State & Input Sets  $X, U$ ]  
Constraints --> ICC[Tightened Constraints:  $h_s \leq x_s + \gamma(\hat{M}_e) \leq h_b$ ]  
Constraints --> Term[Terminal Set  $Z \in \mathcal{M}_e$ ]
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The **System** and **Environment** are decoupled dynamically, but fundamentally linked through perception. The controller cannot redesign the estimator, but it *can* leverage its system-dependent performance.

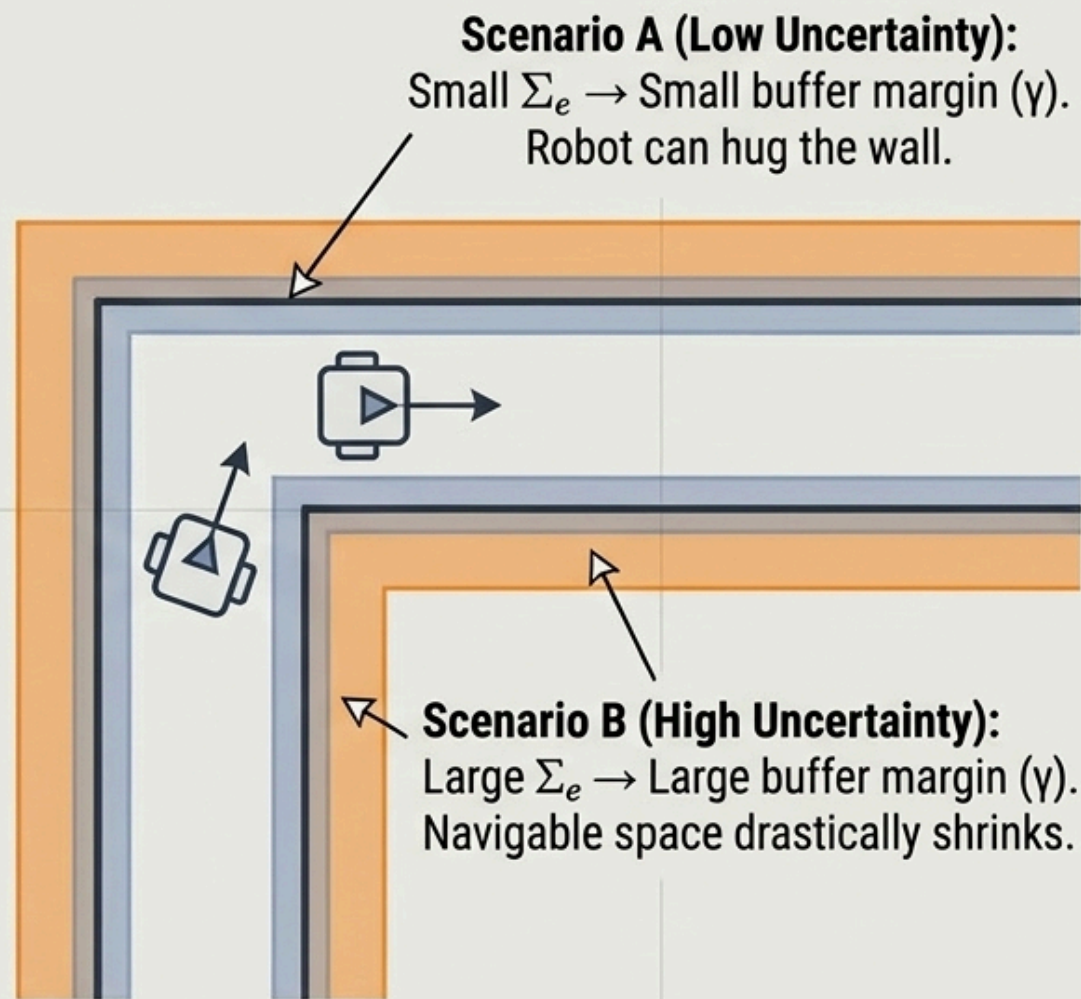
Recursive Feasibility Conditions

Recursive Feasibility Conditions

- Predictor design satisfies monotonic tightening properties relative to updated estimator moments.
- Terminal set $Z_f(M_e)$ is positively invariant under terminal control law κ_f .
- If optimization is feasible at time k , candidate trajectory remains feasible at $k+1$ with high probability.
- Guarantees closed-loop constraint satisfaction under state-dependent feedback coupling.

$$P(h_s x_s + h_e x_e \leq h_b) \geq 1 - \epsilon$$

Enforcing Safety. Uncertainty forces conservative driving. PAC-MPC dynamically tightens constraints based on predicted covariance to guarantee a safe trajectory (e.g., 95% collision avoidance).



Closed-Loop Stochastic Stability

- State norm defined as $||\xi|| = ||x_s|| + ||\Sigma_e||_F$ for joint state and covariance stability.
- Optimal value function V_N^* serves as a Lyapunov function bounded by class K_∞ functions.
- Stage costs satisfy decay inequalities under local control Lyapunov function conditions.
- Proves asymptotic closed-loop stability of system state x_s and covariance Σ_e to target equilibria.

$$V_N(x_s, U_k, \Sigma_e, r_k) = [\text{Stage Costs}] + [\text{Terminal Costs}]$$

$$l_c(x_s, u_s)$$

+

$$l_p(\Sigma_e)$$

+

$$F_c(x_s) + F_p(\Sigma_e)$$

System Tracking Cost:

Minimizes physical deviation from the target state.

Perception Stage Cost:

Penalizes high environment uncertainty, preventing the system from going blind.

Terminal Costs:

Guarantees stability at the end of the prediction horizon.

Unlike standard MPC, PAC-MPC simultaneously minimizes physical tracking error *and* environmental blindness.

Constructive Design for Linear Systems

- Specializes design to linear systems $x_{\{s,k+1\}} = A_s x_{\{s,k\}} + B_s u_{\{s,k\}}$ with polyhedral bounds.
- Terminal set synthesized using Maximum Constrained Admissible Sets (MCAS) O_{∞} with backoff margin η .
- Perception terminal cost F_p formulated over horizon N_p using matrix power terms $\rho^h \|\Lambda^h \bar{\Sigma}_e \Lambda'^h\|_F^2$.
- Solves Linear Matrix Inequalities (LMIs) to guarantee cost decay conditions and stability weights.

Pillar 1: Recursive Feasibility (Theorem 1)

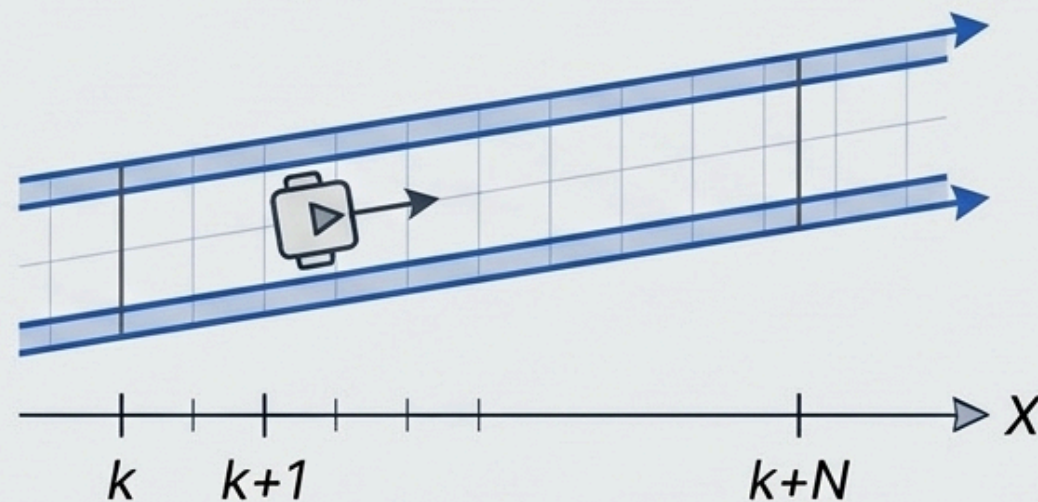
IF the predictor does not underestimate constraint tightening (Assumptions 3 & 4):

$$\gamma(\hat{M}_e) \geq \gamma(M_e)$$

AND

an initial safe state exists

THEN the Terminal Set (Z_f)
ensures a mathematically feasible
path always exists in the future.



Simulation Benchmarks & Performance

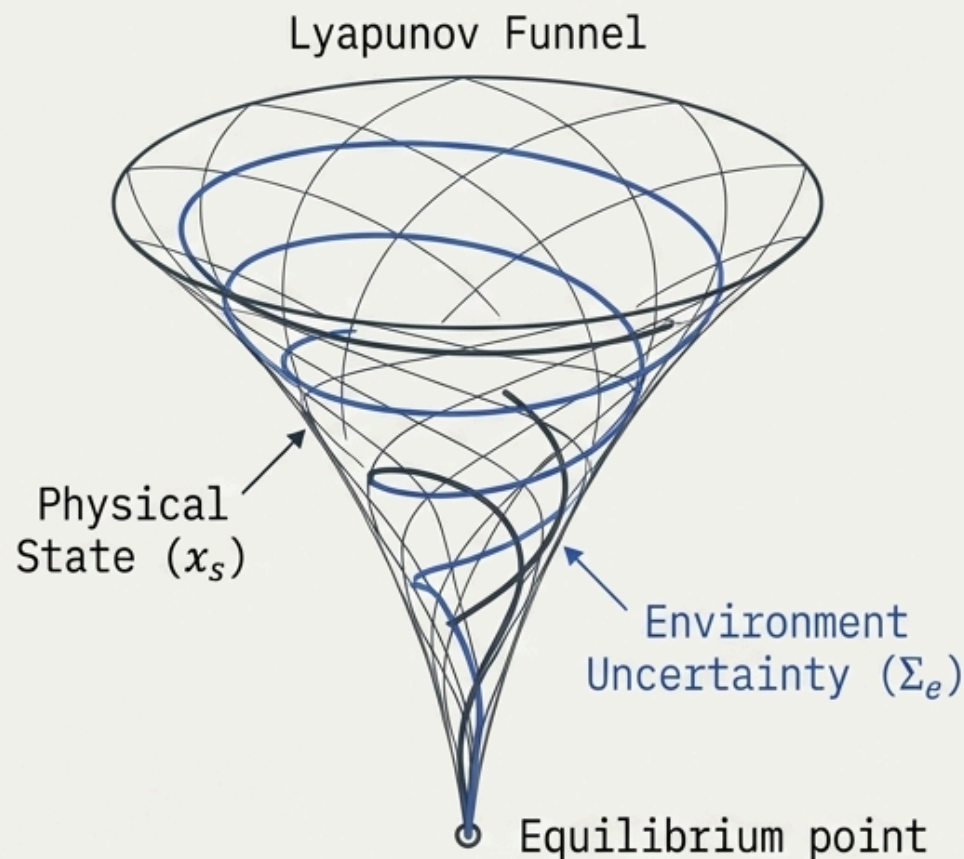
- Evaluated on a double integrator system with input-dependent and state-dependent sensing noise models.
- 100 Monte Carlo runs achieved 98% empirical constraint satisfaction, exceeding the 95% design threshold ($\epsilon = 0.05$).
- Active perception inputs reduced covariance backoff, enabling less conservative closed-loop trajectories.
- Unoptimized Matlab/CasADi/IPOPT solver execution averaged 22 ms per step (< 30 ms peak).

Metric / Parameter	Value / Result	Significance
Target Violation Prob (ϵ)	0.05 (95% safety lower bound)	Design parameter for individual chance constraints
Monte Carlo Constraint Satisfaction	98% across 100 runs (200 trials)	Exceeds theoretical safety bound in closed-loop simulation

Pillar 2: Asymptotic Stability (Theorem 2)

(Condition):

Requires a terminal control cost that bounds perception uncertainty (Assumptions 5 & 6).

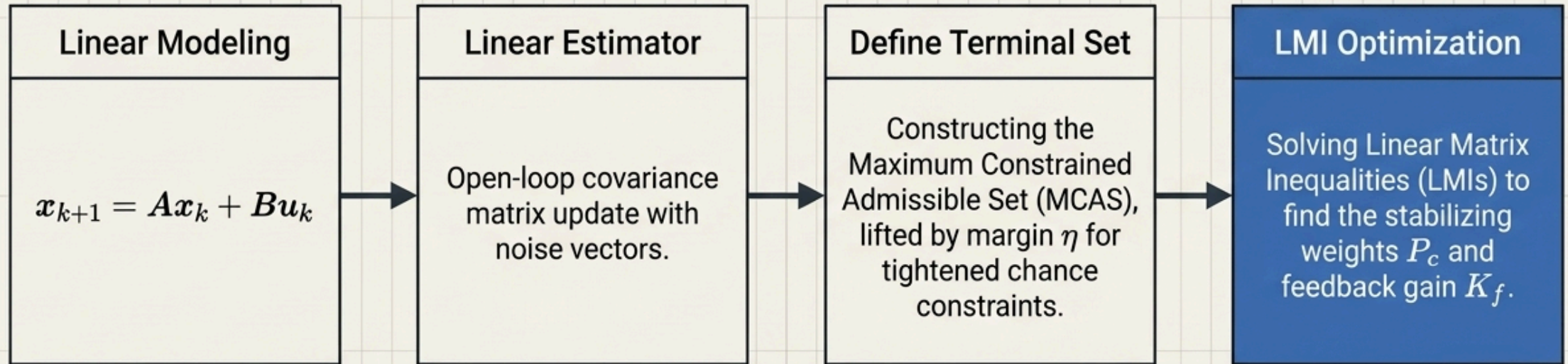


(Guarantee):

Asymptotic Stability of the Augmented State $\xi = (x_s, \Sigma_e)$.

PAC-MPC guarantees that the physical trajectory and the sensory clarity converge simultaneously. The system stabilizes its position while successfully stabilizing its vision.

Constructive Design Pipeline for Linear Dynamics



→ **Constructive Design Pipeline.** For linear systems, the dense nonlinear assumptions of PAC-MPC can be systematically resolved offline using standard LMI optimization, enabling real-time deployment.

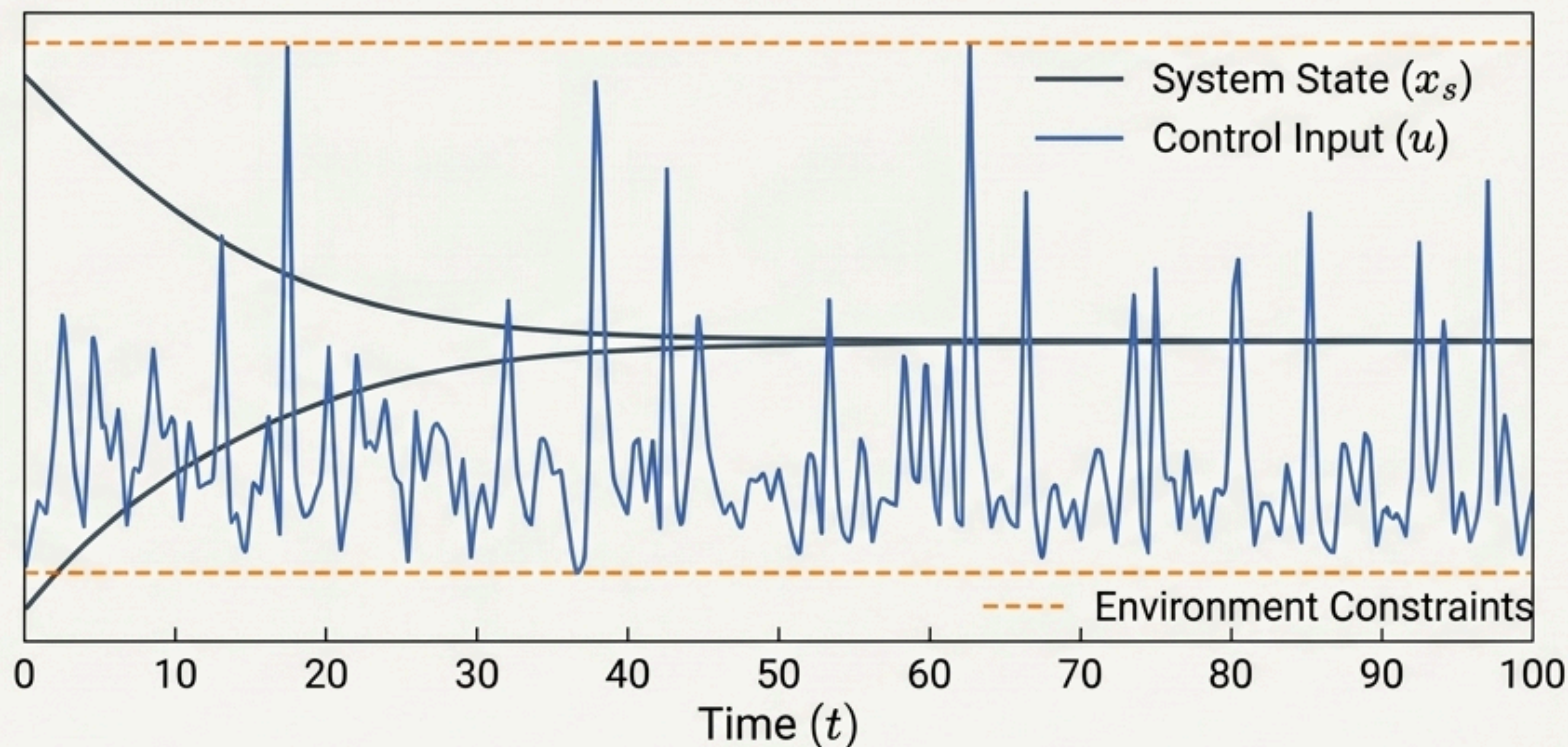
Case Study: Input-Dependent Perception

Setup

Measurement noise strictly decreases as control input u increases.

$$D_e = (1 - \beta[u]_2) \bar{D}$$

The Result



Input-Dependent Perception: The system actively expends control energy solely to improve sensor quality, stabilizing uncertainty without altering its physical destination.

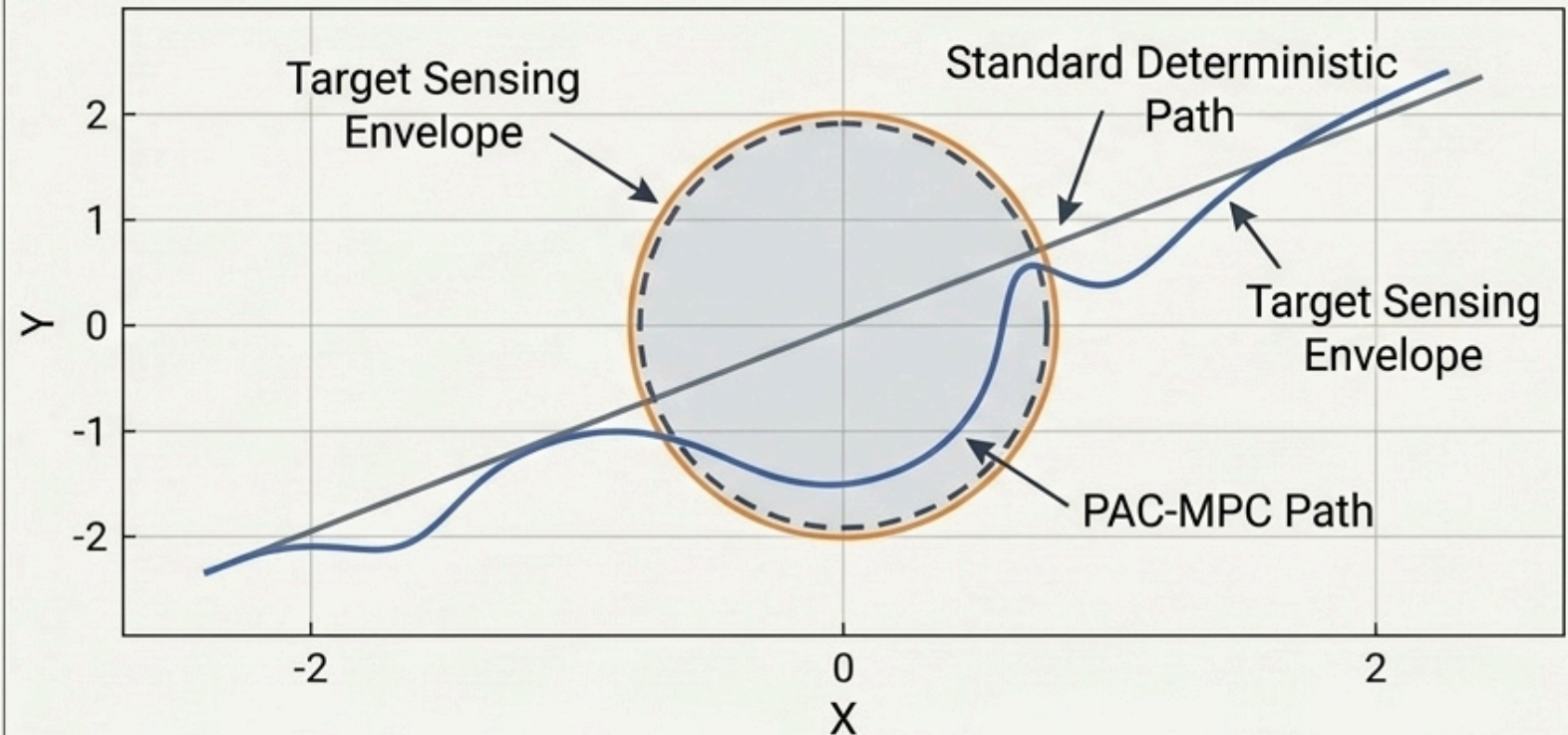
Case Study: State-Dependent Perception

Setup

Measurement quality degrades as distance from target increases.

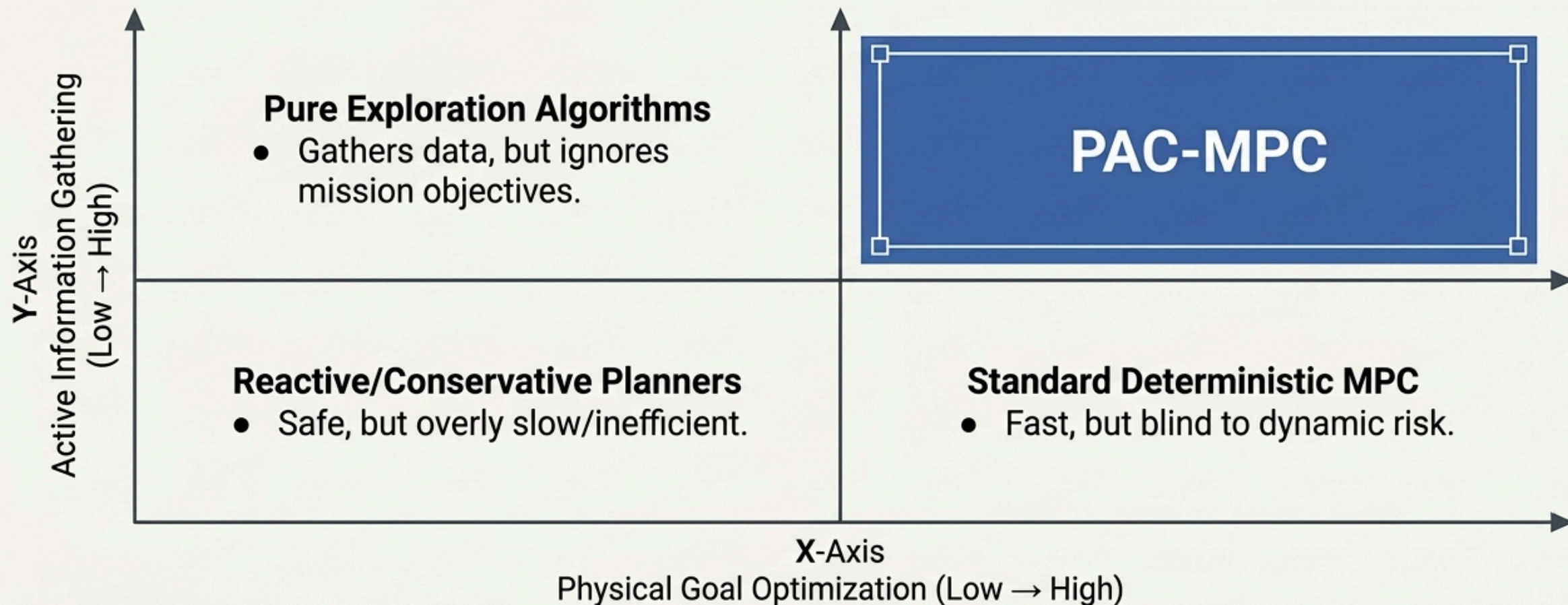
$$[D_e] \propto ([\mu_e] - [x_s])^2$$

The Result



State-Dependent Perception: The system deliberately alters its physical trajectory to prevent field-of-view degradation, taking a longer but fundamentally safer path.

The Synthesis: Surviving the Unknown



By explicitly predicting the interplay between physical action and sensory clarity, PAC-MPC simultaneously optimizes for mission completion and environmental awareness—yielding safe, non-conservative autonomy.

Key Takeaways

- PAC-MPC integrates system dynamics and action-dependent perception quality into a chance-constrained MPC framework.
- Provides formal probabilistic recursive feasibility and closed-loop stochastic stability guarantees for system state and perception uncertainty.
- Delivers real-time constructive LMI design (< 30 ms/step) with superior empirical constraint safety (98%+ satisfaction).

Keywords: Perception-Aware Control, Model Predictive Control, Chance Constraints, Uncertain Environments, Autonomous Systems, Recursive Feasibility, Stochastic Stability

Thank You

Questions & Discussion